

Project Title
DNA Sequence Analysis Tool

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

GeneScan is a DNA sequence analysis tool designed to efficiently search for specific DNA patterns or gene motifs. It uses the Boyer–Moore string matching algorithm to perform faster pattern searching in large DNA sequences. The tool allows users to upload DNA sequences, find matching patterns, highlight matched regions, and view search time and character comparisons. Its performance can also be compared with the Naïve String Matching algorithm. The project demonstrates the practical use of advanced string-matching techniques in bioinformatics and genetic research.

Signature of the Guide